GIT CHEAT SHEET

Terminal only (prompt $ or %). Never the R Console (>). Habit: branch, status, add, commit, push

1. LOOK FIRST (safe)

`git status`	branch + staged/unstaged/untracked
`git branch`	list branches, * = where I am
`git log --oneline -5`	last 5 commits
`git diff`	unstaged line changes
`git diff --staged`	what is about to be committed
`git remote -v`	https:// = GCM, git@ = SSH

Fact: git was written in about 10 days in 2005 by Linus Torvalds.

2. SAVE LOOP

`git add <file>`	put it in the box (stage)
`git commit -m "msg"`	seal + label it (local snapshot)
`git push`	ship the box to GitHub
`git push -u origin  `	first push of a new branch

add without commit = nothing to push. Stage by name, never add -A.

3. BRANCHES

`git checkout -b name`	create + switch
`git switch -c name`	newer form, same thing
`git checkout main`	switch to existing branch
`git branch -d name`	delete merged branch (safe)
`git branch -D name`	force delete (careful)

A branch is a tiny file holding one commit ID. Make them freely. Uncommitted edits follow you to the new branch.

4. MERGING

`git checkout main`	go to the RECEIVING branch
`git merge a3`	bring a3 into main
`git merge --abort`	cancel a scary merge
`git add f ; git commit`	finish a conflict fix

Fast-forward = silent. Merge commit = vim opens. Conflict = delete <<<<<<< ======= >>>>>>> markers.

5. GITHUB SYNC

`git clone <url>`	download a repo (once)
`git clone -b a1 <url> dir`	clone one branch
`git fetch origin`	peek at GitHub, no changes
`git pull`	fetch + merge
`git pull origin main --allow-unrelated-histories`	join repos with separate origins
`git remote set-url origin <url>`	repoint origin
`git remote add origin <url>`	first setup only

origin is just a nickname. 'fetch first' = GitHub has commits I lack: pull, then push.

6. UNDO AND RESCUE

`git restore <file>`	PERMANENT: discard edits
`git restore --staged <f>`	unstage, keep edits
`git reset`	unstage all, files untouched
`git checkout -f  `	DESTRUCTIVE: back up first
`git stash / git stash pop`	shelve / restore edits
`git reflog`	history of HEAD, the rescue tool
`git rm <file>`	delete + stage deletion

Committed work is very hard to lose (reflog keeps ~90 days). Uncommitted work is not.

7. SETUP (one-time)

`git config --global --get-all credential.helper`	want only the GCM path
`git credential-manager configure`	register GCM
`git config --global credential.useHttpPath true`	per-repo creds
`git config --global core.editor nano`	swap out vim

8. VIM ESCAPE

`Esc :wq Return`	save and finish
`Esc :q! Return`	quit, discard

Exiting vim is a top Stack Overflow question. You are in good company.

9. ERROR DECODER

`fetch first`	GitHub ahead: git pull, then push
`non-fast-forward`	git pull --no-rebase origin main
`remote origin already exists`	use set-url, not add
`Already up to date (merge)`	wrong branch: checkout main first
`stuck > or function>`	wrong window. Ctrl+C
`unexpected symbol`	git typed in R Console

DAILY WORKFLOW

`1. git branch, git status`	where am I? clean?
`2. edit files`	
`3. git diff`	what changed?
`4. git add <file>, git status`	right files staged?
`5. git commit -m "..."`	save
`6. git push`	send